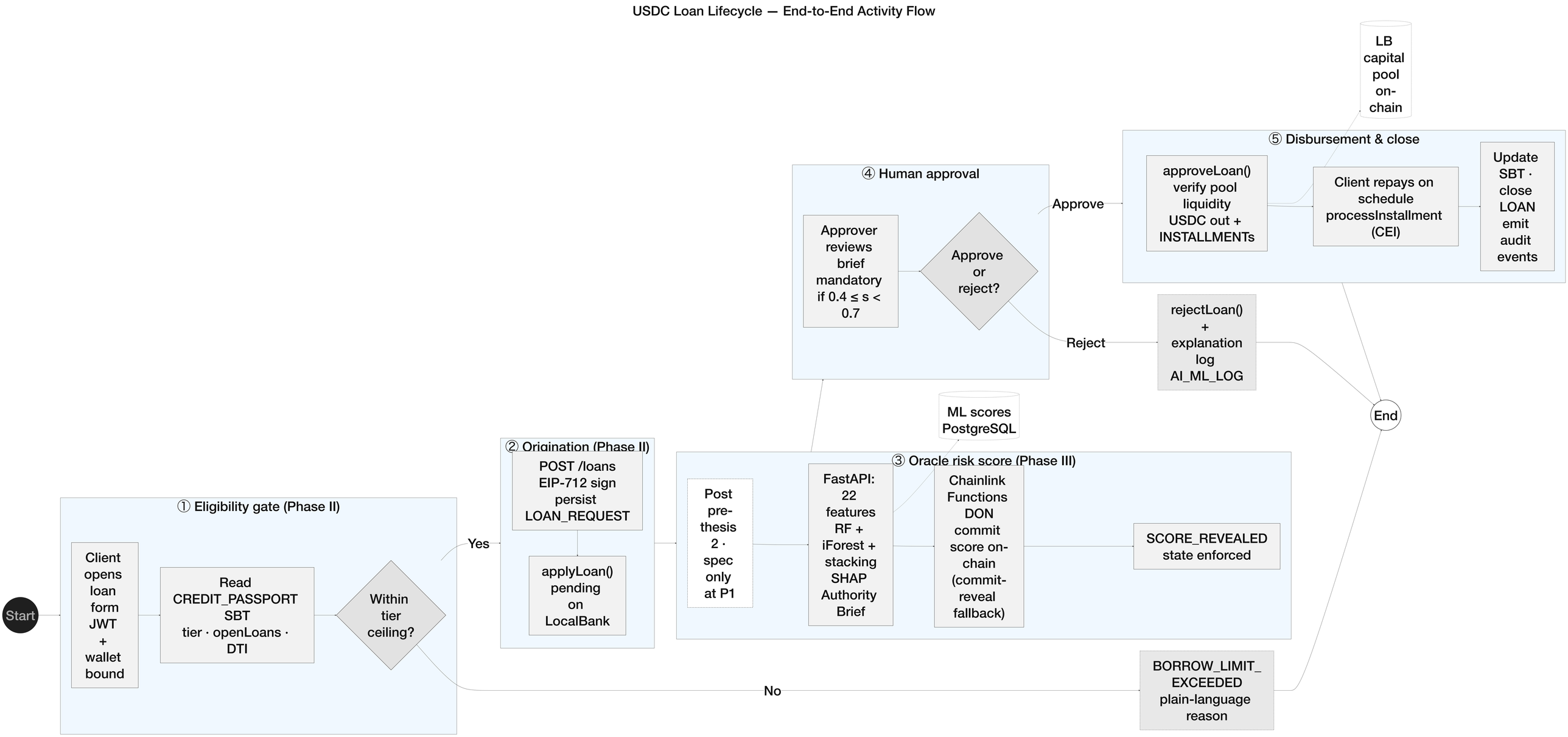
USDC Loan Lifecycle — End-to-End Activity Flow
Start
① Eligibility gate (Phase II)
Client opens loan form JWT + wallet bound
Read CREDIT_PASSPORT SBT tier · openLoans · DTI
Within tier ceiling?
Yes
No
② Origination (Phase II)
POST /loans EIP-712 sign persist LOAN_REQUEST
applyLoan() pending on LocalBank
③ Oracle risk score (Phase III)
Post pre-thesis 2 · spec only at P1
FastAPI: 22 features RF + iForest + stacking SHAP Authority Brief
Chainlink Functions DON commit score on-chain (commit-reveal fallback)
ML scores PostgreSQL
SCORE_REVEALED state enforced
④ Human approval
Approver reviews brief mandatory if 0.4 ≤ s < 0.7
Approve or reject?
Approve
Reject
⑤ Disbursement & close
approveLoan() verify pool liquidity USDC out + INSTALLMENTs
Client repays on schedule processInstallment (CEI)
Update SBT · close LOAN emit audit events
LB capital pool on-chain
rejectLoan() + explanation log AI_ML_LOG
BORROW_LIMIT_EXCEEDED plain-language reason
End